SPRING 2024

Notebook/Assignment No. 02

ZOO518P- Applied Entomology and Pest Management (Practical)

Solved by M.Kashif (03064963933)

Assignment marks: 10

Assignment Questions/Activities:

1. **Collect and identify two insect pests from fruits and vegetables. 6**

1. Fruit Fly (Bactrocera zonata):

- Belongs to the Tephritidae family, which is known for infesting various fruits.
- Adults have a distinctive appearance with a brownish-yellow body, transparent wings with a brown band, and red eyes.



- **Wing pattern:** Transparent wings with a prominent dark brown band along the wing margin.
- **Color**: Brownish-yellow body with dark markings on the thorax.
  Larvae are white, legless, and can be found feeding inside the fruit, causing softening and rotting.
- **Larval appearance**: Creamy-white, elongated, and tapered at both ends. They lack legs and have a pointed mouth hook.

2. Brinjal Fruit and Shoot Borer (Leucinodes orbonalis):

- Belongs to the Crambidae family, which includes many agricultural pest species.

- Adults are small moths with white wings that have brown patterns and a slender body.

- **Wing pattern:** White wings with distinctive brown patterns, including a marginal band and a few spots.
- **Color**: White wings with brown markings, and a brownish-gray body.
  Larvae are pale pinkish-white with a dark head and can be found boring into the shoots and fruits of brinjal (eggplant), causing wilting and fruit damage.
- **Larval appearance:** Pinkish-white, slender, and slightly curved. They have a dark brown head capsule and possess chewing mouthparts.



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Three biological reasons for their identification of Insect Pests (Optional):

1. Morphological characteristics:

- Fruit flies and brinjal fruit and shoot borers have distinct morphological features that aid in their identification, such as wing patterns, body colors, and larval appearance.
- These characteristics help in distinguishing them from other insect species and identifying them accurately.

2. Life cycle and behavior:

- ✓ Understanding the life cycle and behavior of these pests is crucial for their identification and management.
- ✓ Fruit flies lay eggs inside the fruit, and the larvae feed on the fruit pulp, while brinjal fruit and shoot borers lay eggs on the plant surface, and the larvae bore into the shoots and fruits.
- ✓ Knowing these specific behaviors helps in identifying the pest based on the damage symptoms observed on the plant.

3. Host plant specificity:

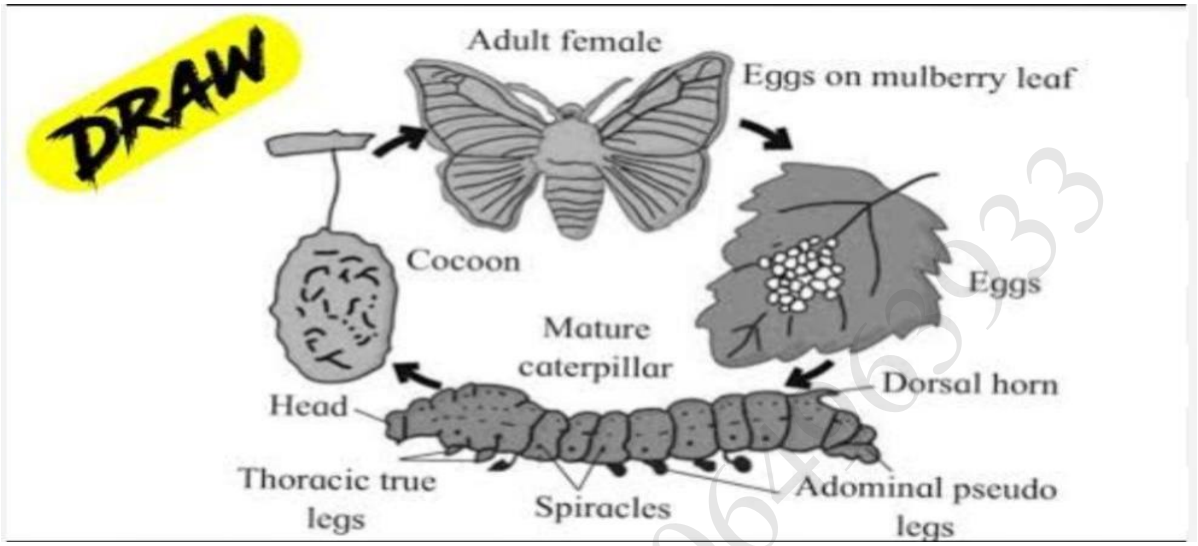
- ✓ Fruit flies have a wide host range and can infest various fruit crops, while brinjal fruit and shoot borers are more specific to brinjal (eggplant).
- ✓ Identifying the host plant and the associated damage symptoms can aid in the identification of the pest species.
- ✓ For example, if the damage is observed on brinjal with wilted shoots and damaged fruits, it is likely caused by the brinjal fruit and shoot borer.

The addition of wing pattern, color, and larval appearance provides more detailed morphological characteristics that can further assist in the accurate identification of fruit flies and brinjal fruit and shoot borers. These specific features, along with the other biological

by M.Kashif(03064963933

aspects mentioned earlier, contribute to the reliable identification of these pest species in Pakistani agricultural settings.

2. Draw the life cycle of silk worm and upload its image on LMS. 2



3. What are the benefits of collecting and preserving insects in museums? 2

Collecting and preserving insects in museums offers several significant benefits. Here's a brief note on the importance of this practice:

1. Scientific research and study:

- Preserved insect specimens serve as valuable resources for scientific research and study.
- Researchers can examine the morphological characteristics, genetic material, and other biological aspects of the specimens to gain insights into their taxonomy, evolution, and ecology.
- Museums provide access to a wide range of insect species, including rare or extinct ones, which may not be available for study otherwise.

2. Biodiversity documentation and conservation:

- Insect collections in museums help document the biodiversity of a region over time.
- By preserving specimens from different periods, museums create a historical record of insect populations and distributions.

by M Kashif 03064463933

- This information is crucial for understanding changes in biodiversity, tracking the impact of environmental factors, and developing conservation strategies for threatened or endangered species.

3. Education and public outreach:

- Museum collections serve as educational tools for students, researchers, and the general public.
- Visitors can observe and learn about the fascinating world of insects through well-curated exhibits and displays.
- Museums often organize educational programs, workshops, and events to raise awareness about the importance of insects in ecosystems and their roles in nature.

4. Identification and pest management:

- Preserved insect specimens act as reference materials for accurate identification of species.
- Researchers, entomologists, and pest management professionals can compare collected specimens with museum collections to identify pests accurately and develop targeted control strategies.
- Museums also provide information on the biology, ecology, and distribution of pest species, aiding in effective pest management approaches.

5. Historical and cultural significance:

- Insect collections in museums often have historical and cultural significance.
- Some collections may include specimens gathered by notable entomologists or explorers, providing insights into the history of entomology and scientific expeditions.
- Insect specimens may also be associated with specific cultural practices, folklore, or traditional knowledge, serving as a means to preserve and understand the cultural heritage related to insects.

In summary, collecting and preserving insects in museums is crucial for scientific research, biodiversity documentation, conservation efforts, education, pest management, and understanding the historical and cultural significance of insects. These collections serve as invaluable resources for researchers, students, and the public, fostering a deeper appreciation and knowledge of the incredible diversity and importance of insects in our world.

by M.Kashif(03064963933

Submission guidelines

1. The due date of assignment submission is **10-07-2024 (Wednesday).**
2. This is an in-line submission format and you have to type your answer in the LMS.
3. You must submit pictures of insects with their scientific names that you collected from fruits and vegetables.
4. Email submission is highly discouraged and is not allowed in any case.
5. If you have any queries, please contact via ticketing service.

by M.Kashif(03064963933

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